

Operating Systems – EDA093/DIT401

Course Introduction

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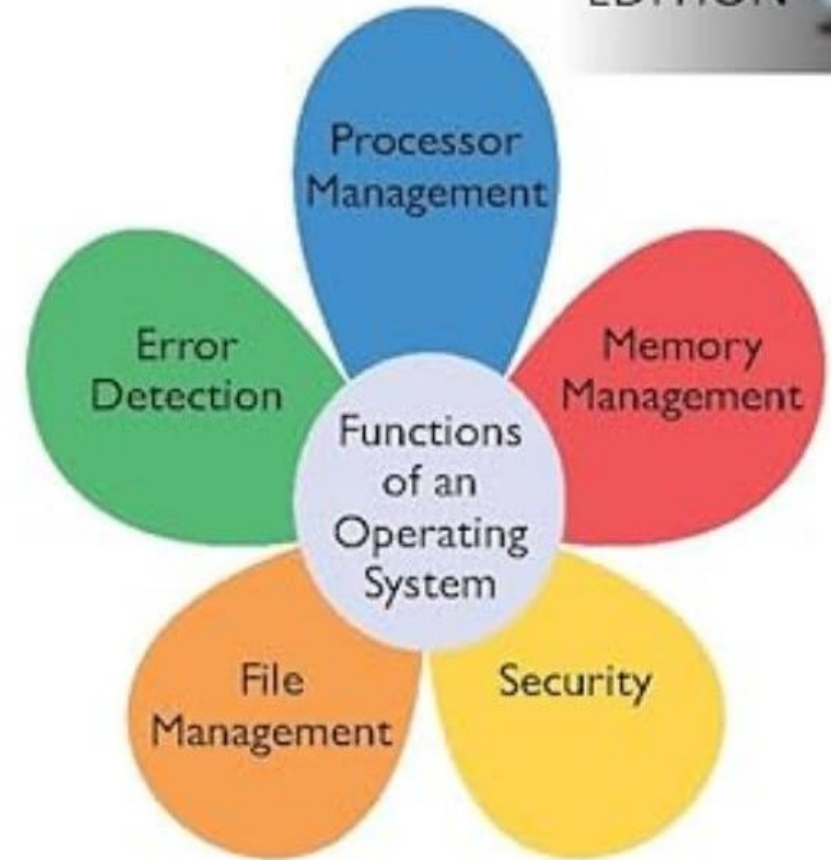
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Agenda

- Main textbook
- Why study Operating Systems?
- What is this course about?
 - What is this course NOT about?
- Best practices

Main textbook

Modern Operating Systems by
Andrew S. Tanenbaum
(fifth edition)



Modern Operating Systems

FIFTH EDITION

Tanenbaum • Bos



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ROBOCOP



```

COMMAND.COM
LOAD BIOS
BIOS SYSTEM CHECK
RAM CHECK
CONFIG.SYS
BIO.COM INTERFACE
TO ROM I/O
CONTROLLER
COMSPEC.EXE
MEMORY.DAT
ROBO UTILS
SYSTEM BUFFER
PARAMETERS
PARITY SET
MEMORY SET
SYSTEM STATUS
OK_
    
```

```

#85  A8 8A
#87  B9 E1 60
#8A  99 F8 83
#8D  8C
#8E  18 F7
#90  AD F3 83
#93  49 A5
#95  8D F4 83
    
```

```

#98  A9 83
#9A  A8 F8
    
```

```

29 SLOTA EQU $06
30 SLOT EQU $07
31 BANK EQU $08
32 COUNT EQU $F9
33 VIEW EQU $FA
34 VIEWM EQU $FB
35 STOR EQU $FC
36 STORM EQU $FD
37 D8 EQU $FE
38 D8H EQU $FF
    
```

```

ADDRESS ADDR (
<<< POKE RAMCARD
<<< POKE 16K RAM
HIBYTE # OF POIN
PAGE VIEWED (IM
HIBYTE OF VIEW
PAGE IN STORAGE
HIBYTE OF STOR
USED FOR BANK S
HIBYTE OF D8-B
    
```

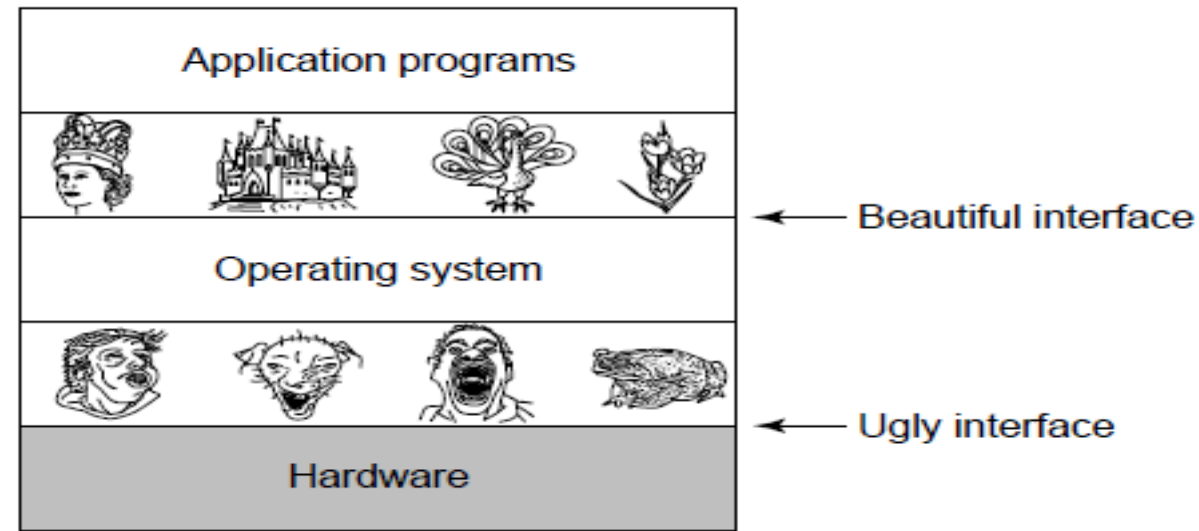
Why study Operating Systems?

Where there are computers, there are operating systems...



Why study Operating Systems?

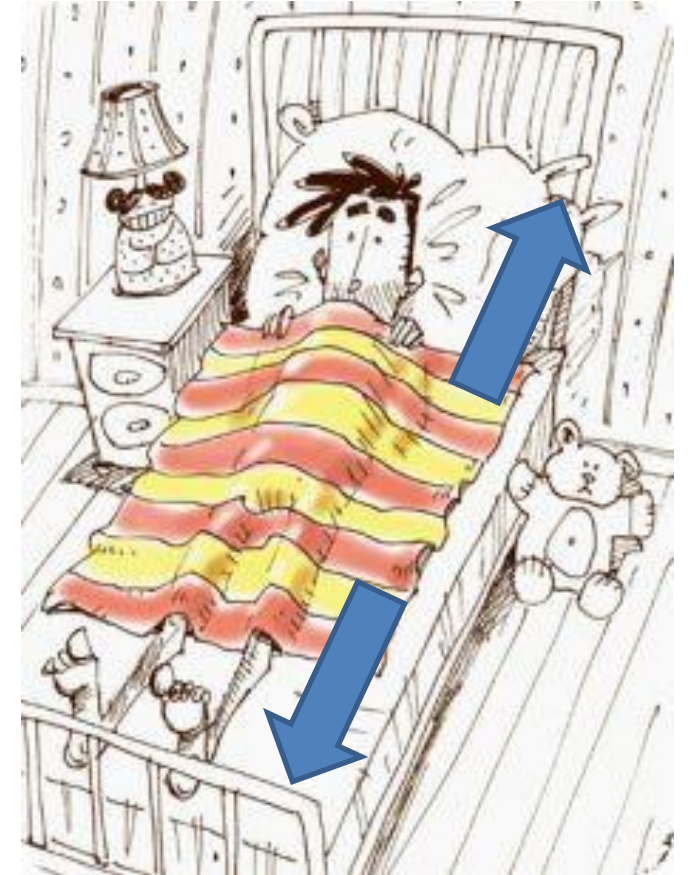
- Provides services to system users
- “Shields” the user from the hardware
- Resource manager:
 - CPU(s)
 - memory
 - I/O devices
 -



Picture source: Modern OS, by A. Tanenbaum

Why study Operating Systems?

- To learn about system functionality and design
 - performance vs. simplicity, HW vs. SW, etc.
- To learn how to manage complexity through appropriate abstractions
- again: Because computers and OSs are everywhere!



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What is this course not about?

What is this course not about?

- Learn how to use Windows
- Learn how to use Linux
- Learn how to use OS X
- Learn how to program in C



What is this course about, then?

Content of the course:

- Introduction / System Structures
- Processes / Threads
- Multithreaded Programming
- Process scheduling
- Synchronization
- Deadlocks
- Memory Management
- Virtual Memory
- File Systems
- I/O Systems
- Security / Protection
- Virtualization

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Best Practices

1. Participate to lectures, exercise sessions and labs
 1. Ask when something is not clear
 2. Ask when you have some doubt about what is being discussed
2. Study at pace with classes, do the labs in time
3. Use the book and other material
 1. Read the book
 2. Try the exercises in the book
4. Be nice to future generations
 1. Provide feedback
 2. Evaluate the course!!!
5. Check the updates in the Canvas page
6. Look at messages in Canvas
7. **FAMILIARIZE YOURSELF WITH UNIX – WITH GIT – WITH GITHUB**

Best Practices

- Do you know what this is?
 - ls
 - mv *filename1 filename2*
 - chmod *options filename*
 - man *commandname*
 - ls -l | grep "Aug" | sort -n
- Install a virtual machine (e.g., Virtual Box) → Install Unix OS (e.g., Ubuntu) → start playing with commands and system calls

Thank you for your attention!

Questions?